PassFrame tuition

Volume & 3D Shapes

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 24cm³

2. 27cm³

3. C — both A (30cm³) and B (30cm³)

4. 12

5. 5

6. 36cm³

7. C — cone

8. 64cm³

9. A, B, C and E (a cuboid has 6 rectangular faces, 12 edges, 8 vertices, and no curved faces)

10. 16cm³

11. 9

12. B — triangular prism

13. 5

14. 18cm³

15. 6

Standard

1. 3cm

2. 1 litre

3. D — neither (A = 40cm³ exactly and B = 36cm³ are both not more than 40cm³)

4. 28cm³

5. 3 litres

6. A — 6 squares arranged in a cross shape

7. 3000cm³ = 3 litres

8. A, B, C and E (volumes 40cm³, 36cm³, 48cm³ and 45cm³ are all between 30 and 50; D=20cm³ is not)

9. B — 27cm³ is greater than A's 24cm³

10. 4cm

11. 4cm

12. A — 3×4×2=24cm³ matches the 6×2×2=24cm³ cuboid (B=32cm³ does not)

13. 42cm³

14. 2500cm³

15. 50cm³

Standard / Hard

1. 20×15×10=3000cm³; a third = 1000cm³

2. Height = 3cm; 4 boxes fit (480 ÷ 120)

3. Edge = 6cm; surface area = 6 × (6×6) = 216cm²

4. 40 m³; 40,000 litres

5. 5cm

6. A, B and D (volume = 72cm³, 8 vertices, 6 faces; it has no curved surface)

7. 30×20×10 = 6000cm³

8. 6×6×3 = 108cm³

9. 5cm (125 ÷ 25)

10. 0.5m³ = 500 litres; 5 minutes

11. 6 cubes

12. 8cm × 6cm × 2cm each

13. Width 5cm, length 10cm (since 10 × 5 × 4 = 200)

14. Container = 1800cm³; each bottle = 125cm³; 1800 ÷ 125 = 14 bottles, remainder 50cm³

15. √(15×12×20) = √3600 = 60cm³

Homework — Answers

Easy / Standard

1. 20cm³

2. 8cm³

3. C — both A (24cm³) and B (24cm³)

4. 12

5. 6

6. 28cm³

7. A — cylinder

8. 125cm³

9. A, B, C and E (a cube has 6 equal square faces, 12 edges, 8 vertices, and no curved faces)

10. 18cm³

11. 8

12. B — square-based pyramid

13. 5

14. 18cm³

15. 5

Standard

1. 3cm

2. 2 litres

3. D — neither (A=36cm³ and B=36cm³ are both exactly 36cm³, not more)

4. 32cm³

5. 4 litres

6. A — 2 triangles and 3 rectangles

7. 5000cm³ = 5 litres

8. A, B, C and E (volumes 48cm³, 50cm³, 48cm³ and 54cm³ are all between 40 and 60; D=20cm³ is not)

9. B — 24cm³ is greater than A's 20cm³

10. 3cm

11. 5cm

12. A — 4×5×2=40cm³ matches the 5×4×2=40cm³ cuboid (B=32cm³ does not)

13. 96cm³

14. 2200cm³

15. 60cm³

Standard / Advanced

1. Height = 6cm (3 times 2cm), so volume = 4 × 2 × 6 = 48cm³

2. B — 125cm³ (edge = 5cm, and 5×5×5=125)

3. Cube's volume = 27cm³, so height = 27 ÷ (9×3) = 1cm

4. A, B and C (width = 4cm, height = 4cm, so volume = 12×4×4 = 192cm³)

5. Cuboid's volume = 64cm³, so the cube's edge = 4cm (since 4×4×4=64)

6. 30×20×10=6000cm³ = 6 litres

7. 6×4×4 = 96cm³

8. Edge = 5cm; surface area = 6 × 25 = 150cm²

9. B — 3cm

10. Total volume = 40cm³; stack height = 10cm

11. A, B, C and E (all give a volume of 36cm³; D gives 40cm³)

12. 9cm × 4cm × 1.5cm each

13. Width 4cm, length 8cm (since 8 × 4 × 4 = 128)

14. Container = 1920cm³; each bottle = 64cm³; 1920 ÷ 64 = 30 bottles exactly, 0 left over

15. √(20×15×12) = √3600 = 60cm³